Dissertation on

A Deep Learning and Generative AI-Driven Diet Recommendation System for Personalized Chronic Disease Management

Submitted in partial fulfillment of the requirements for

Bachelors in Technology in

Computer Science and Engineering

UE23CS320A - Capstone Project Phase - 1

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CERTIFICATE

This is to clarify that the dissertation entitled

A Deep Learning and Generative AI-Driven Diet Recommendation System for Personalized Chronic Disease Management

is a bonafide work carried out by

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In partial fulfillment for the completion of Fifth-semester Capstone Project Phase - 1 (UE23CS320A) in the Program of Study - Bachelor of Technology in Computer Science and Engineering under rules and regulations of PES University, Bengaluru during the period Aug 2025 – Dec. 2025. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report. The dissertation has been approved as it satisfies the 5th-semester academic requirements in respect of project work.

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DECLARATION

We hereby declare that the Capstone Project Phase - 1 entitled **"A Deep Learning and Generative AI-Driven Diet Recommendation System for Personalized Chronic Disease Management"** has been carried out by us under the guidance of **Dr Saritha K, Dept. of Computer Science and Engineering** and submitted in partial fulfillment of the course requirements for the award of the degree of **Bachelor of Technology** in **Computer Science and Engineering of PES University, Bengaluru** during the academic semester Aug. – Dec 2025. The matter embodied in this report has not been submitted to any other university or institution for the award of any degree.

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ACKNOWLEDGEMENT

I would like to express my gratitude to **Dr. Saritha K**, Department of Computer Science and Engineering, PES University, for his/ her continuous guidance, assistance, and encouragement throughout the development of this UE23CS320A - Capstone Project Phase – 1.

I am grateful to all Capstone Project Coordinators, for organizing, managing, and helping with the entire process.

I take this opportunity to thank **Dr. Sandesh B J**, Professor & Chairperson, Department of Computer Science and Engineering, PES University, for all the knowledge and support I have received from the department. I would like to thank **Dr. B.K. Keshavan**, Dean of Faculty, PES University for his help.

I am deeply grateful to Late **Dr. M. R. Doreswamy**, Founder, PES University, whose vision and dedication continue to inspire generations of learners. I would also like to express my sincere gratitude to **Prof. Jawahar Doreswamy**, Chancellor, PES University, **Dr. Suryaprasad J**, Vice-Chancellor, PES University and **Prof. Nagarjuna Sadineni**, Pro Vice-Chancellor, PES University for providing to me various opportunities and enlightenment every step of the way. Finally, this project could not have been completed without the continual support and encouragement I have received from my family and friends.

ABSTRACT

Patients with chronic diseases like Type 2 Diabetes Mellitus , Chronic Kidney Disease, and Hypertension need to adhere to a strict diet throughout their lives to ensure a smooth life ahead. Conventional digital tools that are not very user-friendly and simply do not respond to the continuous changing daily lifestyle of the patients and their changing nutritional needs. The project proposes a Deep Learning and Generative AI-Driven Diet Recommendation System that will deliver personalized, adaptive, and medically compliant diet plans to chronic disease patients. The system considers patient-specific parameters including BMI, age, weight, comorbidities, and dietary preferences to design an ideal meal plan via a hybrid model that combines Deep Learning, Generative AI, and nutritional rule-based constraints. A feedback loop is made where the patient can record changes to the initial plan which results in meals being recalibrated in real-time.. The system provides an optional supervision option that allows someone to monitor the patient. This allows the patients or their family members to grant access to doctors or any helper for supervising the patient's progress and monitoring any alerts in case of emergencies. There is a chatbot integrated into the system that helps by providing disease-specific information and assistance.

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CHAPTER 1

INTRODUCTION

1.1 Background

Chronic diseases like Type 2 Diabetes Mellitus , Chronic Kidney Disease and Hypertension are dependent on a controlled diet. Patients have trouble adhering to regular diet charts as diet schedules are usually static, generic, and are not inclusive of day-to-day lifestyle changes. Furthermore, diet personalization is usually lacking in mobile applications, and are often disregarding medical parameters which leads to disengagement over time.

1.2 Need and Motivation

Generative AI and Deep Learning have improved to the point where diet plans can now be personalized taking into account the user's health profile, user preferences, and daily activity. Recommendations can be tailored to the user's age, BMI, and conditions as well as dietary preferences. The ability to create real time personalized diet plans is a mainly unexplored field and is a strong motivation behind this work. An additional optional system to alert a personal doctor/supervisor, in case of critical events is integrated.

1.3 Project Overview

Deep learning and Generative AI technology will inform systems in this project to offer personalized and adaptive meal plans tailored to specific diseases and conditions. The system will adjust nutrition re-calculation based on the user reporting eating deviation/s and will ensure nutritional intake for the remaining day is safe and balanced for the given condition. Relative to the user, an integrated, supportive AI system will be able to provide instructions and basic information on the disease. Based on the combination of automation and human input, the system will provide user-centric, practical, and automated solutions to facilitate long-term management of diets in chronic diseases.

CHAPTER 2

PROBLEM STATEMENT

2.1 Core Problem

Chronic illnesses such as T2 Diabetes, Chronic Kidney Disease, and Hypertension all require extremely specific and ongoing dietary restraint, and yet, the majority of patients find it very difficult to comply with those restraints, if they can do so at all. Current dietary plans available are often unchanging, one-size-fits all, and do not take into account the daily fluctuations of the patients activities and food consumption. Consequently, patients do not achieve critical dietary goals, and the likelihood of complications increases, leading to adverse overall outcomes.

2.2 Limitations and Current Solutions

Today, most mobile applications and nutritional tools do not offer complete personalization of the plans created, and tend to overlook key medical parameters such as comorbidities, sodium restrictions, and carbohydrate quotas. Also, there are no available systems that manage, in real time, deviations from the prescribed plan, such as patients eating food that they are not allowed to eat. These systems do not protect patients from harmful dietary mistakes, as this recalculation is not done automatically, nor are any medical restrictions built into these systems. Finally, previous assessments have noted these systems do not have safety layers, for example, if a patient is fully ignoring any restrictions these systems impose, there are no recommendations at all.

2.3 Need for a Smarter, Safer System

An intelligent solution is needed that can create personalized, adaptive, and medically aligned meal plans, and adapts to users' actions/activities in real time. Furthermore, the system should offer supervision mode functionality, enabling doctors or approved caregivers to observe patient compliance when needed. Closing these gaps offers a more reliable and feasible dietary management solution for chronic disease patients.

CHAPTER 3

LITERATURE SURVEY

3.1 Past Paper References

3.1.1 AI Nutrition Recommendation Using a Deep Generative Model and ChatGPT (2024)

- Methodology: Deep Generative Network (VAE + RNN/GRU) for user meal generation . The optimizer layer ensures zero calorie deviation and LLM's like chatgpt expand the database
- Results: Achieved 0.00% calorie error and 87% macronutrient accuracy. Used Explainable AI for clustering
- Limitations**:** Proof-of-concept only it lacks long-term user validation.

3.1.2 AI-Driven Personalized Nutrition (RAG-Based Digital Health Solution) (2025)

- Methodology: Retrieval-Augmented Generation (RAG) framework using "Virtual Nutritionist" for T2DM & obesity.
- Key Results**:** 80%+ recipe adherence to dietary guidelines; 92% sustainability compliance.
- Limitations**:** Focused on limited food domain (smoothies); not full-scale chronic disease management.

3.1.3 Integrated AI Framework for Personalized Nutrition (2025)

- Methodology: Hybrid ML + LLM framework with natural-language user inputs. Predicts energy and macronutrient needs; creates meal plans via rule-based filtering.
- Key Results: Improved calorie/macronutrient predictions using user-level inputs; natural interface for personalization.
- Limitations: Doesn't include explicit disease biomarkers or medical constraints.

3.1.4 ChatDiet: LLM-Augmented Food Recommender Chatbot (2024)

- Methodology: LLM-Augmented chatbot with orchestration layer combining personal & population models.
- Key Results: 92% user satisfaction and adherence in pilot test; effective personalization and interactivity.
- Limitations: Not specialized for chronic diseases; limited clinical evaluation.

3.1.5 NUTRIVISION: Automatic Diet Management in Smart Healthcare (2024)

- Methodology: Combines computer vision (Faster-R-CNN) for food recognition + ML for personalized NCD diet management.
- Key Results: Fast, automated nutritional estimation from food images; applied to diabetes & heart disease datasets.
- Limitations: Image recognition inaccuracies; lacks adaptive feedback loop or generative meal synthesis.

3.1.6 Artificial Intelligence in Personalized Nutrition and Food

- Methodolog: Comprehensive review of DL, RL, FL for dynamic nutrition frameworks.
- Key Results: Establishes AI as transformative for chronic disease diet personalization; highlights federated learning for privacy.
- Limitations: Conceptual review; limited empirical or generative AI evaluation.

3.1.7 Using LLMs for Nutrition Planning in CKD (2025)

- Methodology: LLM-based assistance in interpreting and customizing CKD diet plans.
- Key Results: Demonstrates LLMs can interpret complex diet data to support kidney patients.
- Limitations: Conceptual; lacks concrete ML pipeline or quantitative validation.

3.1.8 Balanced Diet Recommendation Using Fuzzy Logic (2023)

- Methodology: Fuzzy Sugeno model using basic demographic inputs (age, height, weight).
- Key Results: 86.5% accuracy in macronutrient balance calculation.
- Limitations: Simplistic, no chronic disease or AI personalization; rule-based only.

CHAPTER 4

RESEARCH GAP

4.1 Limited Chronic Disease Personalization

The currently available solutions (Papastratis et al., Aydın et al., ChatDiet) primarily target individuals with general dietary needs or those suffering with one comorbidity, namely diabetes or obesity. Despite the fact that numerous patients have multiple comorbidities, there still is no single unified AI system that deals with personalized diets for T2DM, CKD, and Hypertension simultaneously.

4.2 Weak Medical Constraint Enforcement

AI generative diets (Papastratis; Gavai; Kenger) frequently create dishes that seem valid and plausible; however, these do not necessarily adhere to certain medical nutrient restrictions such as the limits for sodium, potassium, phosphorus, and carbohydrates. A system that integrates AI with stringent medical rule-checking is required.

4.3 No Real-Time Adaptation After User Deviations

None of the reviewed works recalculate or adjust the remainder of the day's diet when a patient consumes something that is not a part of the prescribed meal plan. Chronic disease care frequently involves the possibility of problematic deviations, and thus the lack of a system that accounts for such is a major gap.

4.4 No Supervision or Accountability Mechanism

Despite the fact that numerous chronic patients need oversight for safety, existing tools do not provide an option for doctor or caregiver monitoring. The flexibility and oversight tools that enable regulatory supervision is not documented in current literature.

4.5 Lack of Clinical Validation and Long-Term Studies

Most systems (Papastratis, ChatDiet, NUTRIVISION) are based on simulated data or have been tested on a short timescale. The use of AI in chronic disease populations is still in the early stages, and there is little in the way of robust real-world evaluation of such diet systems.

4.6 Poor Integration of Chatbot + Diet Generation + Safety Layer

Some studies explore chatbots, others look at diet creation, and others at health advice—but no one integrates all three:

- AI diet creator
- Teaching chatbot
- Safety / oversight

This is detrimental to overall chronic disease management.

4.7 Insufficient Consideration of Cultural or Localized Dietary Patterns

A number of these works examine and create Western style meals (e.g Mediterranean, DASH) with no regard to surrounding food cultures. The lack of focus on an Indian or culturally specific eating pattern greatly decreases its uptake potential.

4.8 Absence of Change Over Time Learning and Behaviour Tracking

Current systems/programs do not account for the patient's behaviour over time, for example, repeated changes to the diet, improved, or worsened health. There is no chronic adaptive system that is behaviour based.

CHAPTER 5

OBJECTIVES

5.1 Primary Objective

Develop a personalized device and mobile App in Primary Health Care (PHC) settings that uses Deep Learning and Generative AI to design and implement personalized, flexible, and clinically-appropriate diet recommendations for patients with chronic conditions including T2DM, CKD and Hypertension.

5.2 Functional Objectives

Use Generative AI and NHS dietary recommendations to create a pre-set menu for specific diseases.

A user-friendly chronic disease dietary management system that allows constant diet recommendation adjustments in real-time. The App should provide a chatbot for educational and advisory purposes including basic user-robot interaction.

Allow user system input of physical data (height, weight, gender, and age) and health profile (diseases, medications, and allergies) along with food preferences.

5.3 Safety and Monitoring Objectives

Maintain unbreached medical nutrient thresholds e.g., sodium, potassium, phosphorus, carbohydrates.

Integration of a supervision option to allow a clinician or a trusted relative to monitor the user's adherence to the diet recommendations.

Observe and notify the user of potential risky behavior or excessive non-compliances.

5.4 System-Level Objectives

Adhere to evidence-based dietary instructions.

Adhere to a user-friendly, flexible, and expandable design to enable incorporation of management for other chronic diseases or dietary approaches.

Focus the system on facilitating chronic disease self-management and improving user experience.

CHAPTER 6

OVERVIEW OF DATASETS

6.1 List of Datasets

6.1.1 Indian Food Composition Tables (IFCT) 2017 - ICMR-NIN

The Indian Food Composition table is published by Indian Council of Medical Research's National Institute of Nutrition which is located in Hyderabad. It contains information about 528 most commonly eaten Indian food items. It spreads across 151 macronutrients, micronutrients, vitamins, minerals and other dietary components. It gives region based food across different Indian states and accounts for changes in food compositions across the country. The data is available as a pdf on the NIN website

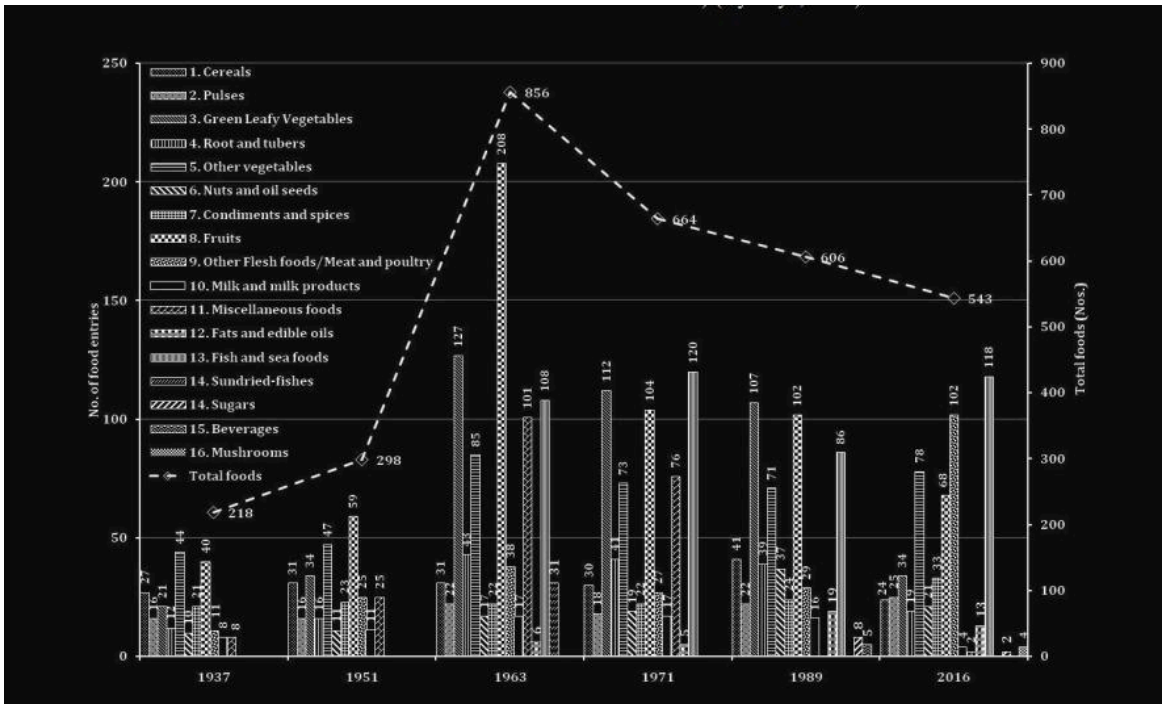

Fig. 1: IFCT Dataset

6.1.2 Kaggle Indian Food 101 Dataset

The Kaggle Indian Food 101 dataset is a community based dataset that contains information on 255 popular Indian dishes and Recipes. It contains completely prepared meals like curries, breads, etc. It contains dish name,ingredients, cooking method,regional origin ,dietary classification ,course type and approximate nutritional information. It provides real world information on what people actually eat.The dataset is available for free on kaggle website and is the bridge between IFCT and meal planning.

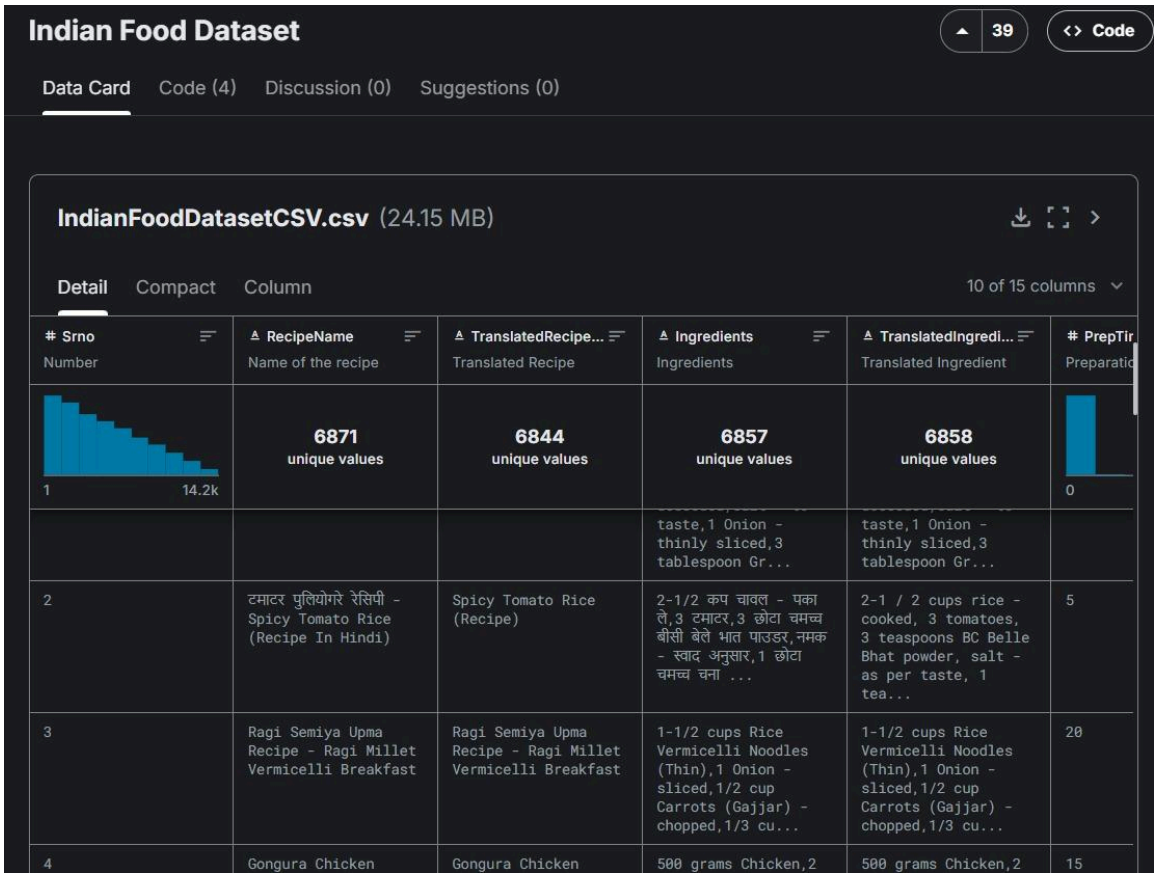

Fig. 2: 101 Indian Food Dataset

6.1.3 Open Food Facts India

Open Food Facts is an open-source database of food products. It has over 50000 packaged food items that are available in India .Each product contains an entry with a barcode making it easy for mobile scanning . We can get information on nutritional facts, additives, allergens and a nutrition-score. The database is crowd sourced and people can always add new products, food is logged. The full database is freely accessible, and due to being licensed as Open Database License (ODbL) it is ideal for commercial and academic work. You can download the entire database in bulk or interact with it through the API.

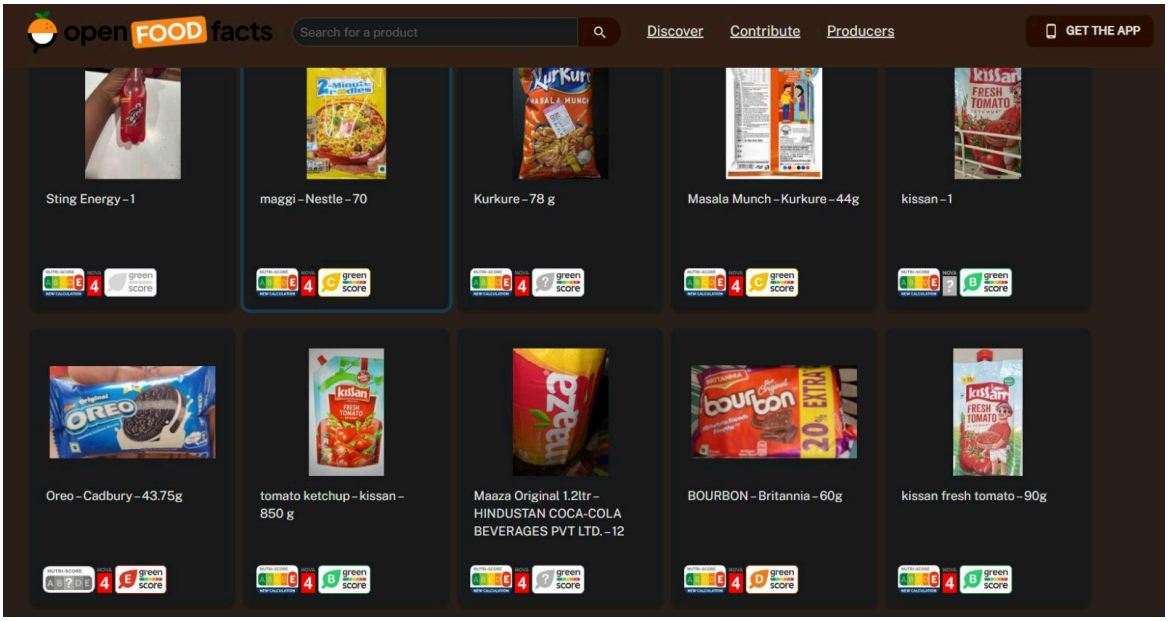

Fig. 3: Open Food Facts India

6.1.4 Synthea: Generation of Synthetic Patients

Synthea is a tool provided by MITRE Corporation and is a reputed tool in information and student research . During research in healthcare informatics, situations arise where real patient data cannot be used due to HIPAA or similar regulations like in our country India . Synthea is able to produce confidential synthetic patient records of varying degrees of medical complexity to be used in records and create a virtual population. The medical records of Synthea patients include a chronic disease history, demographics, and a complete disease history hypertension, chronic kidney disease, diabetes, and even social determinants of health. Every Synthea has a social history including education, income, and a smoking history. If one were to create a complete virtual population, the medical records of each patient would be complete.The ICMR-INDIAB report states that diabetes is at 11.4% in the Indian population. Synthea can generate 1000 patients from which 114 will likely have diabetes based on the appropriate age and BMI distributions. The software is a Java Application and can produce FHIR, HL7 and CSV files which are usable in our models.

CONCLUSION

This project aims to develop a personalized nutrition recommendation system using advanced AI technologies for Indian patients with type 2 diabetes, hypertension, and chronic kidney disease, addressing critical gaps in chronic disease management. Through federated learning architecture, we try to enable a local on device training of models and the updates exchanged will be encrypted, ensuring patient's privacy. The project uses publicly available Indian cuisine databases -IFCT 2017, Kaggle Indian Food Dataset, Open Food Facts to ensure that the diets recommended are in adherence to the indian culture.a

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